

BLAIN VALVES FOR HYDRAULIC ELEVATORS

Excellence in Simplicity and Performance



 **PRODUCT
CATALOGUE**

blain.de

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CONTENT

KV-Series

Mechanical control valve for small lifts

KV1P	- Valve for platform or goods lift	4
KV1S	- Valve for platform or goods lift	5
KV2P	- Valve for goods or home lift	6
KV2S	- Valve for goods or home lift	7

EV-Series

Mechanical control valve for commercial & home lifts

EV0	- Valve for platform or goods lift	8
EV1	- Valve for platform or goods lift	9
EV10	- Valve for home or goods lift	10
EV100	- Valve (fully adjustable) for home lift & commercial lift	11

SEV-Series

Servo electronic valve for wide pressure and temperature range

Excellent ride quality independent of oil temp. & load for commercial & hospital lift	12
---	----

EV40-Series

VVVF control valve for high performance passenger elevators

VVVF Inverter driven, energy efficient control valve for high usage lift	13
--	----

GV

Mechanical control valve for car parking platforms

Simple valve with many applications for car parking lift & goods lift	14
---	----

R10-Series

Rupture valve

Rupture valve (safety valve) in case of free fall due to hose pipe rupture	15
--	----

L-Series

UCM (A3) safety valve against unintended car movement

L10	- Standalone safety valve	16
L20	- Built-on safety valve	17
iL10-S	- Built-in safety valve	18

MRL-H

Machine room less rescue unit

Machine room less rescue system for fast and easy rescue operations	19
---	----

BV

Ball valve

Ball valve for isolating the control valve for servicing and inspection	20
---	----

TH

Tank heater

Tank heater for maintaining oil temperature in cold environment	21
---	----

HP

Hand pump for emergency operations

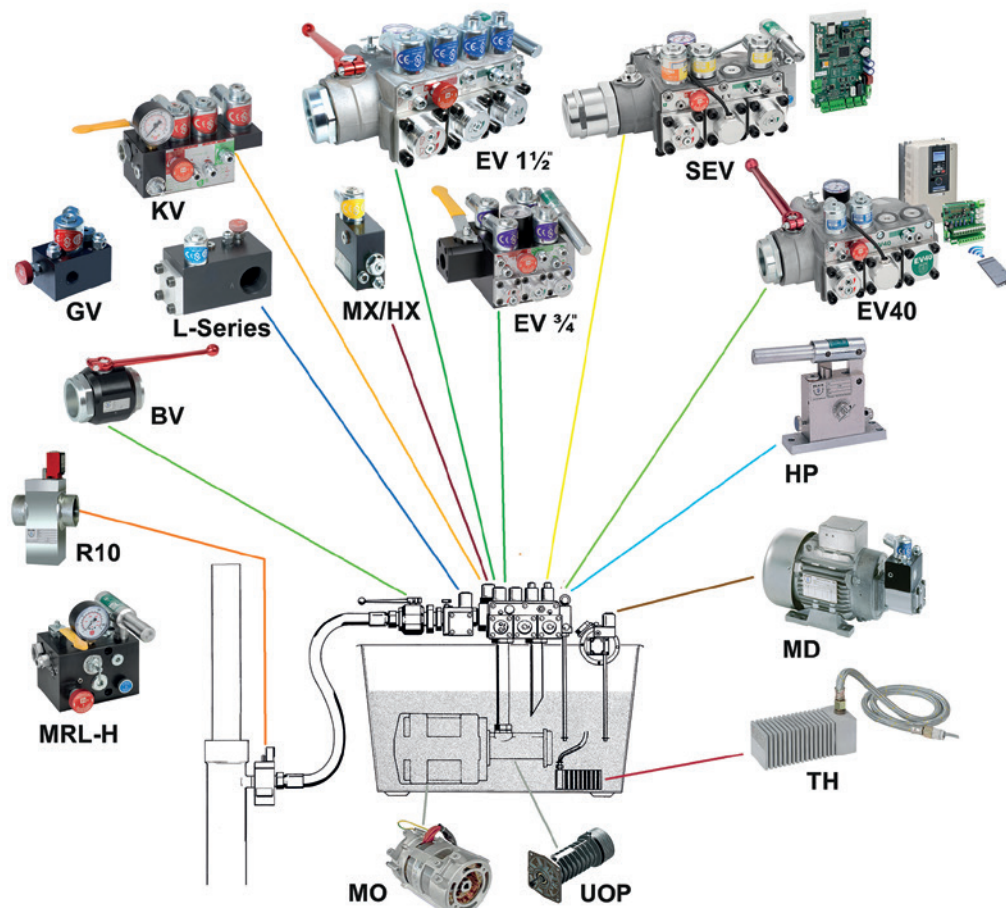
Hand pump to assist in hydraulic lifting	22
--	----

HX-Series

Manual down valve

Extra down speed valve for testing rupture valve	23
--	----

MX-Series	Solenoids down valve	
	Extra down speed valve for testing rupture valve	24
EN	Emergency coil	
	Emergency coil for ARD's (Automatic Rescue Device)	25
KSB	Slack rope valve	
		26
MD	Micro Levelling Drive	
		27
MO	Submersible motor	
		28
UOP	Submersible screw pump	
		29
	Contacts at Blain	
		34





KV1P

Up: One speed
Down: One speed
Max speed: 0.16 m/s (32 fpm)
Max flow: 80 l/min (21 US gpm)

Characteristics

Operation	Medium	Operating pressure	Flow rate
solenoid, electrical	hydraulic oil	8–100 bar (116–1450 psi)	Min: 5 l/min (1.3 US gpm) Max: 80 l/min (21 US gpm)

Oil temperature range: 20°–70°C (68–158°F) - (depending on viscosity grade of oil).

Coil insulation class ~/= IP 68.

Description

KV valves are easy to adjust, compact & simple in design.

KV1P is suitable for platform & goods lifts.

UP direction

The elevator runs with one UP speed up to 0.16 m/s (32 fpm).

The UP start has built-in damping.

The UP stop is caused by de-energizing the motor.

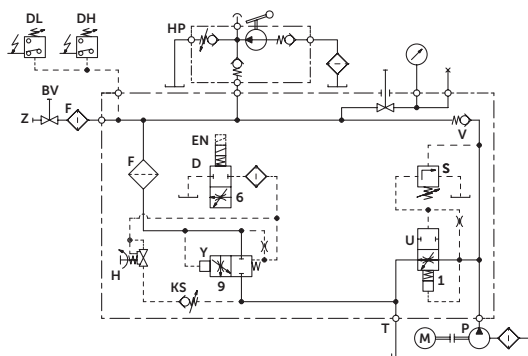
DOWN direction

The elevator runs with one DOWN speed up to 0.16 m/s (32 fpm).

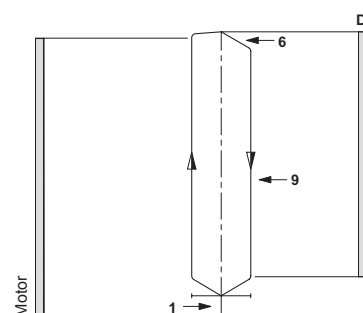
The DOWN start has adjustable damping and the DOWN speed is adjustable.

The DOWN stop has built-in damping.

Hydraulic circuit



Electrical sequence





KV1S

Up: One speed
Down: One speed
Max speed: 0.16 m/s (32 fpm)
Max flow: 80 l/min (21 US gpm)
 with soft stop

Characteristics

Operation	Medium	Operating pressure	Flow rate
solenoid, electrical	hydraulic oil	8–100 bar (116–1450 psi)	Min: 5 l/min (1.3 US gpm) Max: 80 l/min (21 US gpm)

Oil temperature range: 20°–70°C (68–158°F) - (depending on viscosity grade of oil).

Coil insulation class ~/= IP 68.

Description

KV valves are easy to adjust, compact & simple in design.

KV1S is suitable for platform & goods lifts.

UP direction

The elevator runs with one UP speed up to 0.16 m/s (32 fpm) with an adjustable soft stop or up to 0.4 m/s (80 fpm) with overtravel and re-levelling.

The UP start has built-in damping.

The UP stop has adjustable damping (delayed motor stop required).

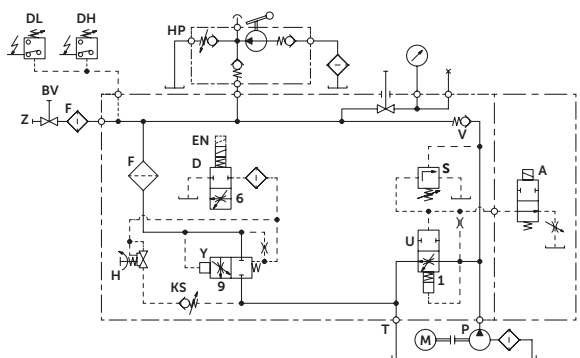
DOWN direction

The elevator runs with one DOWN speed up to 0.16 m/s (32 fpm).

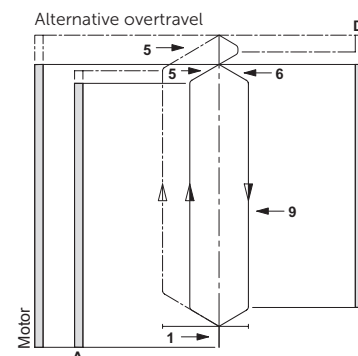
The DOWN start has adjustable damping and the DOWN speed is adjustable.

The DOWN stop has built-in damping.

Hydraulic circuit



Electrical sequence





KV2P

Up	One speed
Down:	Two speeds
Max. lifting speed:	0.16 m/s (32 fpm)
Max. lowering speed:	1 m/s (200 fpm)
Max flow:	80 l/min (21 US gpm)

Characteristics

Operation	Medium	Operating pressure	Flow rate
solenoid, electrical	hydraulic oil	8–100 bar (116–1450 psi)	Min: 5 l/min (1.3 US gpm) Max: 80 l/min (21 US gpm)

Oil temperature range: 20°–70°C (68–158°F) - (depending on viscosity grade of oil).

Coil insulation class ~/= IP 68.

Description

KV valves are easy to adjust, compact & simple in design.

KV2P is suitable for home lifts & goods lifts with two down speeds.

UP direction

The elevator runs with one UP speed up to 0.16 m/s (32 fpm).

The UP start has built-in damping.

The UP stop is caused by de-energizing the motor.

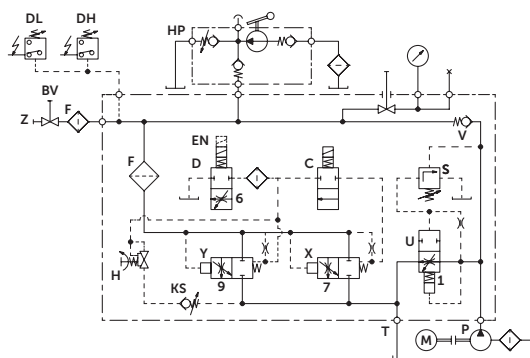
DOWN direction

The elevator runs with two DOWN speeds up to 1 m/s (200 fpm), one full speed and one levelling speed. The DOWN full speed and levelling speed are adjustable.

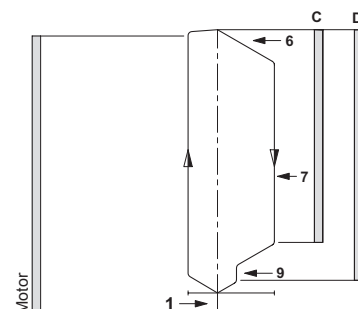
The DOWN start has adjustable damping.

The slow down and DOWN stop have built-in damping.

Hydraulic circuit



Electrical sequence





KV2S

Up:	One speed
Down:	Two speeds
Max. lifting speed:	0.16 m/s (32 fpm)
Max. lowering speed:	1 m/s (200 fpm)
Max flow:	80 l/min (21 US gpm) with soft stop

Characteristics

Operation	Medium	Operating pressure	Flow rate
solenoid, electrical	hydraulic oil	8–100 bar (116–1450 psi)	Min: 5 l/min (1.3 US gpm) Max: 80 l/min (21 US gpm)

Oil temperature range: 20°–70°C (68–158°F) - (depending on viscosity grade of oil).

Coil insulation class ~/= IP 68.

Description

KV valves are easy to adjust, compact & simple in design.

KV2S is suitable for home lifts & goods lifts with two down speeds.

UP direction

The elevator runs with one UP speed up to 0.16 m/s (32 fpm) with an adjustable soft stop or up to 0.4 m/s (80 fpm) with overtravel and re-levelling.

The UP start has built-in damping.

The UP stop has adjustable damping (delayed motor stop required).

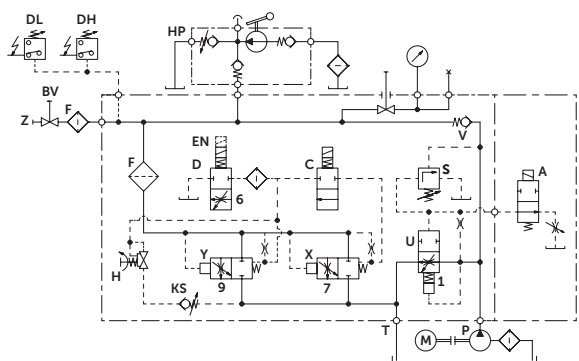
DOWN direction

The elevator runs with two DOWN speeds up to 1 m/s (200 fpm), one full speed and one levelling speed. The DOWN full speed and levelling speed are adjustable.

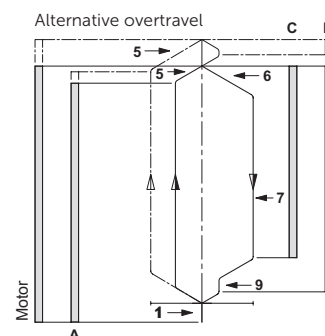
The DOWN start has adjustable damping.

The braking and stopping have built-in damping.

Hydraulic circuit



Electrical sequence





3/4" EVO

10-125 l/min (2-33 US gpm)



1 1/2" & 2" EVO

30-800 l/min (8-208 US gpm)



2 1/2" EVO

500-1530 l/min (130-400 US gpm)

Characteristics

Operation	Medium	Operating pressure	Operating pressure CSA
solenoid,	hydraulic	3/4" 8-100 bar (116-1450 psi)	3/4" 8-100 bar (116-1450 psi)
electrical	oil	1 1/2"/2" 8-100 bar (116-1450 psi)	1 1/2"/2" 8- 70 bar (116-1015 psi)
		2 1/2" 8- 68 bar (116- 986 psi)	2 1/2" 8- 47 bar (116- 690 psi)

Oil temperature range: 20°-70°C (68-158°F) - (depending on viscosity grade of oil).

Coil insulation class ~/= IP 68.

Description

Easy to install, EV's are smooth, reliable and precise in operation throughout extreme load and temperature variations. According to customers' information, valves are factory adjusted ready for operation and very simple to readjust if so desired. The up levelling system combined with compensated pilot control ensure stability of elevator operation and accuracy of stopping. Depending on the flow, available port sizes are 3/4", 1 1/2", 2" and 2 1/2" pipe threads.

UP direction

The elevator runs with one UP speed up to 0.16 m/s (32 fpm).

The UP start is smooth and adjustable.

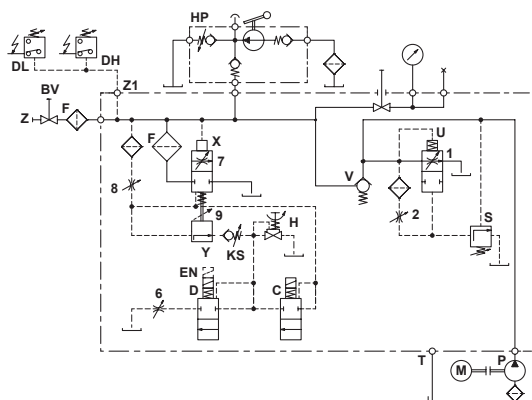
The UP stop is caused by de-energizing the motor.

DOWN direction

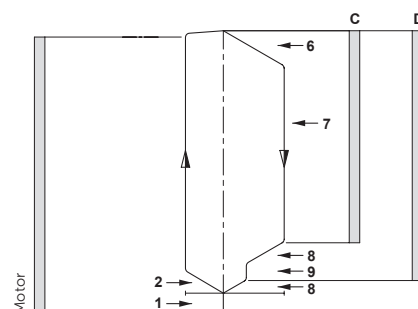
The elevator runs with two DOWN speeds up to 1 m/s (200 fpm), one full speed and one levelling speed.

All DOWN functions are smooth and adjustable.

Hydraulic circuit



Electrical sequence




3/4" EV1

10-125 l/min (2-33 US gpm)


1 1/2" & 2" EV1

30-800 l/min (8-208 US gpm)


2 1/2" EV1

500-1530 l/min (130-400 US gpm)

Characteristics

Operation	Medium	Operating pressure	Operating pressure CSA
solenoid, electrical	hydraulic oil	3/4" 8-100 bar (116-1450 psi)	3/4" 8-100 bar (116-1450 psi)
		1 1/2"/2" 8-100 bar (116-1450 psi)	1 1/2"/2" 8- 70 bar (116-1015 psi)
		2 1/2" 8- 68 bar (116- 986 psi)	2 1/2" 8- 47 bar (116- 690 psi)

Oil temperature range: 20°-70°C (68-158°F) - (depending on viscosity grade of oil).

Coil insulation class ~/= IP 68.

Description

Easy to install, EV's are smooth, reliable and precise in operation throughout extreme load and temperature variations. According to customers' information, valves are factory adjusted ready for operation and very simple to readjust if so desired. The up levelling system combined with compensated pilot control ensure stability of elevator operation and accuracy of stopping. Depending on the flow, available port sizes are 3/4", 1 1/2", 2" and 2 1/2" pipe threads.

UP direction

The elevator runs with one UP speed up to 0.16 m/s (32 fpm) with an adjustable soft stop or up to 0.4 m/s (80 fpm) with overtravel and re-levelling.

The UP start is smooth and adjustable.

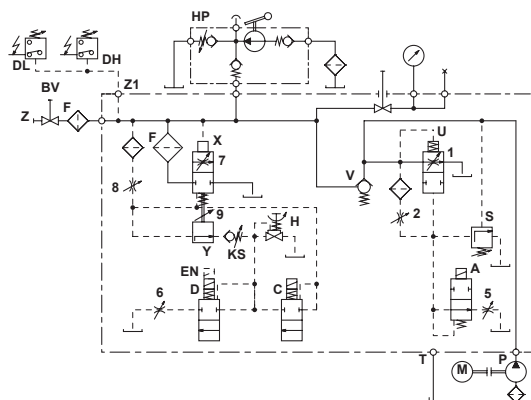
The UP stop is smooth and exact through valve operation, because the motor is running approx. 1 second longer through a time relay.

DOWN direction

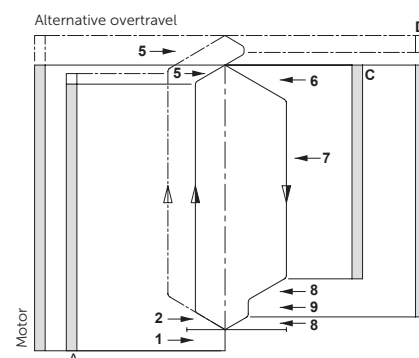
The elevator runs with two DOWN speeds up to 1 m/s (200 fpm), one full speed and one levelling speed.

All DOWN functions are smooth and adjustable.

Hydraulic circuit



Electrical sequence





3/4" EV10

10-125 l/min (2-33 US gpm)



1 1/2" & 2" EV10

30-800 l/min (8-208 US gpm)



2 1/2" EV10

500-1530 l/min (130-400 US gpm)

Characteristics

Operation	Medium	Operating pressure	Operating pressure CSA
solenoid,	hydraulic	3/4" 8-100 bar (116-1450 psi)	3/4" 8-100 bar (116-1450 psi)
electrical	oil	1 1/2"/2" 8-100 bar (116-1450 psi)	1 1/2"/2" 8- 70 bar (116-1015 psi)
		2 1/2" 8- 68 bar (116- 986 psi)	2 1/2" 8- 47 bar (116- 690 psi)

Oil temperature range: 20°-70°C (68-158°F) - (depending on viscosity grade of oil).

Coil insulation class ~/= IP 68.

Description

Easy to install, EV's are smooth, reliable and precise in operation throughout extreme load and temperature variations. According to customers' information, valves are factory adjusted ready for operation and very simple to readjust if so desired. The up levelling system combined with compensated pilot control ensure stability of elevator operation and accuracy of stopping. Depending on the flow, available port sizes are 3/4", 1 1/2", 2" and 2 1/2" pipe threads.

UP direction

The elevator runs with two UP speeds up to 1 m/s (200 fpm), one full speed and one levelling speed.

The UP start and slow down are smooth and adjustable.

The UP levelling speed is adjustable.

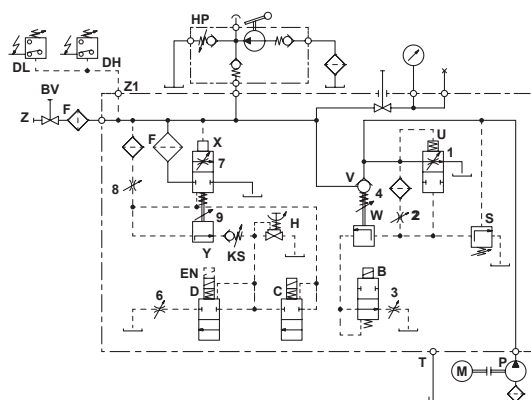
The UP stop is caused by de-energizing the motor.

DOWN direction

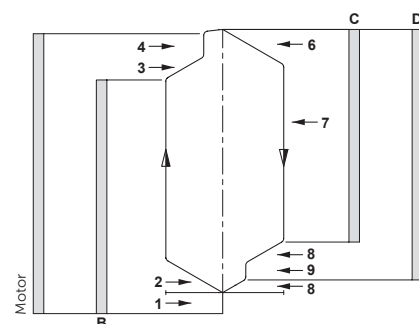
The elevator runs with two DOWN speeds up to 1 m/s (200 fpm), one full speed and one levelling speed.

All DOWN functions are smooth and adjustable.

Hydraulic circuit



Electrical sequence





3/4" EV100

10-125 l/min (2-33 US gpm)



1 1/2" & 2" EV100

30-800 l/min (8-208 US gpm)



2 1/2" EV100

500-1530 l/min (130-400 US gpm)

Characteristics

Operation	Medium	Operating pressure	Operating pressure CSA
solenoid,	hydraulic	3/4" 8-100 bar (116-1450 psi)	3/4" 8-100 bar (116-1450 psi)
electrical	oil	1 1/2"/2" 8-100 bar (116-1450 psi)	1 1/2"/2" 8- 70 bar (116-1015 psi)
		2 1/2" 8- 68 bar (116- 986 psi)	2 1/2" 8- 47 bar (116- 690 psi)

Oil temperature range: 20°-70°C (68-158°F) - (depending on viscosity grade of oil).

Coil insulation class ~/=: IP 68.

Description

Easy to install, EV's are smooth, reliable and precise in operation throughout extreme load and temperature variations. According to customers' information, valves are factory adjusted ready for operation and very simple to readjust if so desired. The up levelling system combined with compensated pilot control ensure stability of elevator operation and accuracy of stopping. Depending on the flow, available port sizes are 3/4", 1 1/2", 2" and 2 1/2" pipe threads.

UP direction

The elevator runs with two UP speeds up to 1 m/s (200 fpm), one full speed and one levelling speed.

All UP functions are smooth and adjustable.

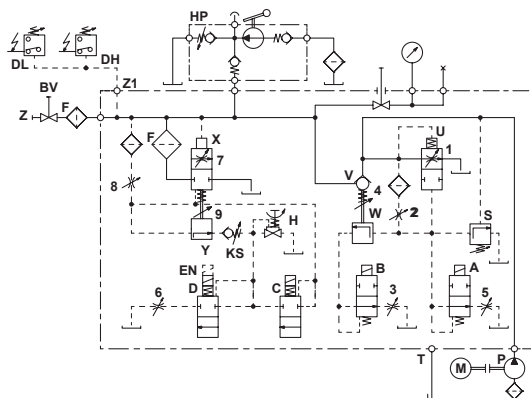
The UP stop is smooth and exact through valve operation, because the motor is running approx. 1 second longer through a time relay.

DOWN direction

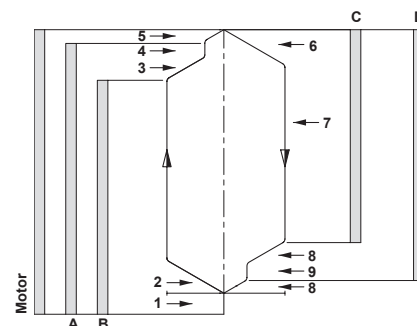
The elevator runs with two DOWN speeds up to 1 m/s (200 fpm), one full speed and one levelling speed.

All DOWN functions are smooth and adjustable.

Hydraulic circuit



Electrical sequence





SEV

40-1200 l/min (10-317 US gpm)

Characteristics

Operation	Medium	Operating pressure	Operating pressure CSA
solenoid, electronic controlled	hydraulic oil	1"-2" 9-100 bar (130-1450 psi) 2½" 9- 68 bar (130- 986 psi)	1"-2" 9-70 bar (130-1015 psi) 2½" 9-47 bar (130- 690 psi)

Oil temperature range: 20°-70°C (68-158°F) - (depending on viscosity grade of oil).
Coil insulation class ~/=: IP 68.

Description

The Servo Electronic Valve (SEV) is available in four different sizes from 1", 1½", 2" and 2½" for a wide range of flow rates. The software stored on the SEV card uses a new sensor system, which measures both pressure and temperature to control ride characteristics of the elevator. It is controlled by closed loop digital electronics, ensuring constant speeds, acceleration and deceleration of hydraulic elevators independent of load and oil temperature. The SEV is easily connected, programmed and adjusted via a Wi-Fi interface using the smartphone. Due to built-in flexibility and simple handling of the user interface, the ride characteristics of the elevator can be adapted to the customer's wishes at any time.

UP direction

The elevator runs with three UP speeds up to 1 m/s (200 fpm), one full speed, one levelling speed and one inspection speed.

All UP transitions are smoothly programmable.

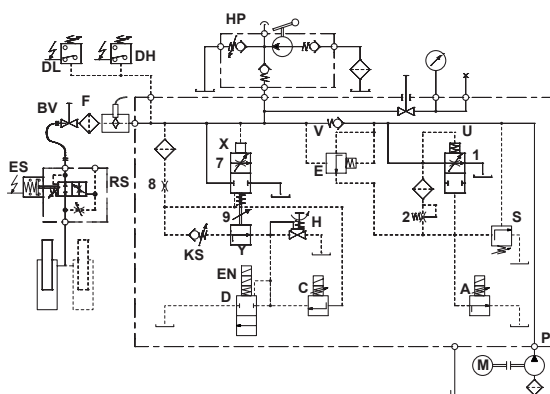
The UP stop is smooth and exact through valve operation, because the motor is running approx. 1 second longer through a time relay.

DOWN direction

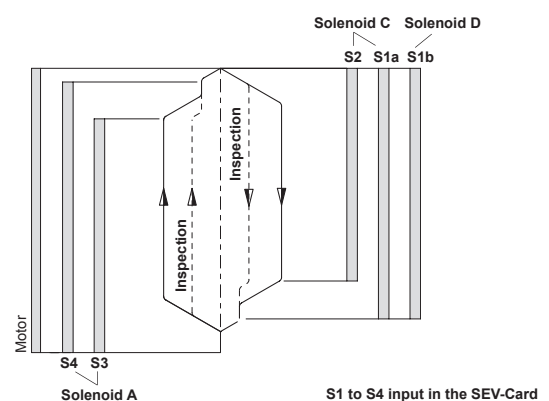
The elevator runs with three DOWN speeds up to 1 m/s (200 fpm), one full speed, one levelling speed and one inspection speed.

All DOWN transitions are smoothly programmable.

Hydraulic circuit



Electrical sequence





3/4" EV40
10-125 l/min (2-23 USgpm)



1 1/2" & 2" EV40
30-800 l/min (8-212 USgpm)



2 1/2" EV40
500-1530 l/min (130-405 USgpm)

Characteristics

Operation	Medium	Operating pressure	Operating pressure CSA
solenoid,	hydraulic	3/4" 8-70 bar (116-1015 psi)	8-55 bar (116-797 psi)
electrical	oil	1 1/2"/2" 8-70 bar (116-1015 psi)	8-55 bar (116-797 psi)
		2 1/2" 8-68 bar (116-986 psi)	8-55 bar (116-797 psi)

Oil temperature range: 20°-70°C (68-158°F) - (depending on viscosity grade of oil).

Coil insulation class ~/= IP 68.

Description

The EV40 system is a smart, modern, easy to install and reliable vvvf solution, which achieves up to 65% energy savings and 50% less oil heating. The EV40 system uses the GA700 inverter from Yaskawa to control UP direction travel, while the DOWN direction travel is handled mechanically by the control valve itself. The low heat input ensures, that oil coolers are no longer required. In this way, the EV40 system offers the most cost-effective and energy-efficient solution in the market for high-traffic elevators. The system is ideally suited for frequently used elevators with high energy-saving potential and those with extreme loads and temperature fluctuations. The intuitive operation of the EV40 via smartphone makes the system the perfect solution for modernization using the latest technology.

UP direction

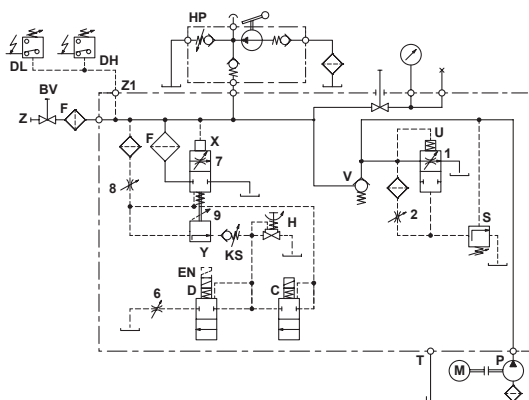
The elevator runs with three fully adjustable speeds for full speed, inspection speed and slow speed.

The start, speeds, transitions and soft stop are customizable parameters, which can be accessed through the menu via smartphone.

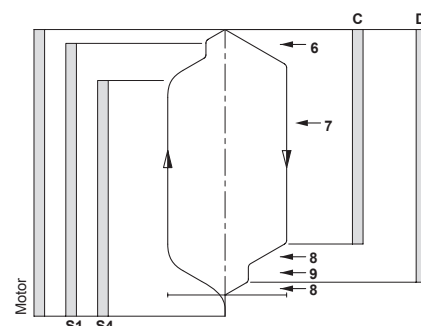
DOWN direction

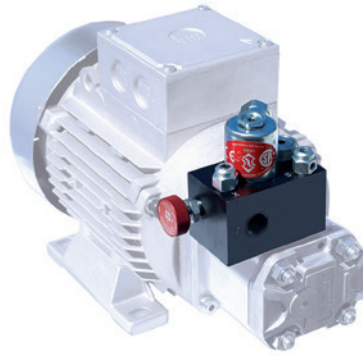
The elevator runs with two DOWN speeds, one full speed and one levelling speed. All DOWN functions are smooth and adjustable.

Hydraulic circuit



Electrical sequence





GV

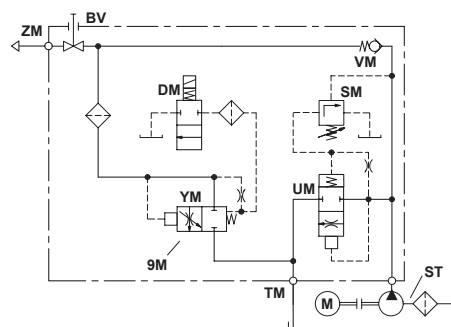
Characteristics

Operation	Medium	Operating pressure	Flow rate
solenoid, electrical	hydraulic oil	3–130 bar (44–1885 psi)	Min: 1 l/min (0.3 US gpm) Max: 24 l/min (6.3 US gpm)
Oil temperature range: 20°–70°C (68–158°F) - (depending on viscosity grade of oil).			
Coil insulation class ~/= IP 68.			

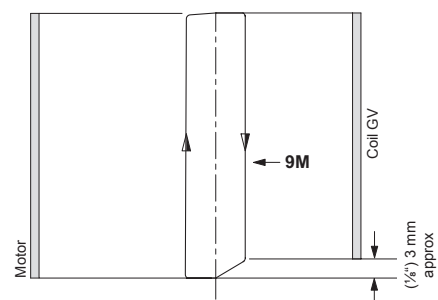
Description

The Blain car parking platform valve GV can be used in car parking applications where the platform needs to be raised above the ground to accommodate another car below the port. Alternatively, this valve is also ideal for lifting material, cargo and suitable for dumbwaiters and goods lifts. The valve offers a single up speed and an adjustable down speed.

Hydraulic circuit



Electrical sequence





R10
up to 2100 l/min (554 US gpm)



R10L
up to 2100 l/min (554 US gpm)



R10+DK+ES
up to 2100 l/min (554 US gpm)

Characteristics

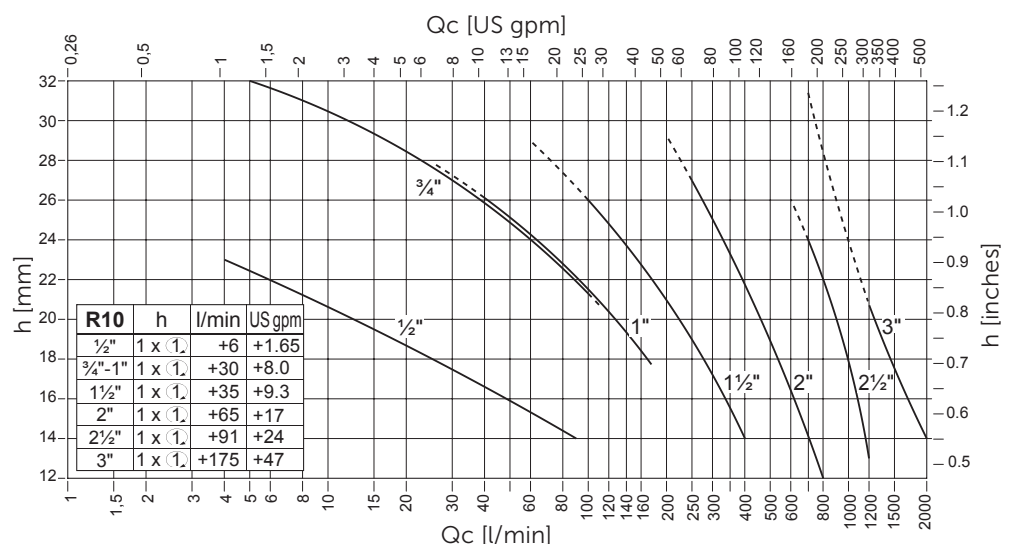
Operation	Medium	Operating pressure	Flow rate
—	hydraulic	1/2"-2" 10–100 bar (950–1350 psi) US 80 bar (145–1160 psi)	Min: 4 l/min (1.1 US gpm)
	oil	2 1/2"-3" 8– 80 bar (690–880 psi) US 47 Bar (680 psi)	Max: 2100 l/min (554 US gpm)

Description

In the event of failure in the main cylinder line due to hose pipe rupture or where the down speed exceeds allowable limits, the R10 valve closes, bringing the car to a smooth stop. Through additional options the closing of the R10 can be electrically signaled (option ES). Synchronized closing of tandem cylinders is also possible (option DK). The connections for the different cylinder and tank ports can be chosen freely. There are inside and outside threads as well as NPT, BSP, metric, Victaulic and flange - connection to choose from.

Range
----- Adjustable
===== Permitted*

*AR 2014/33/EU





1/2" L10
up to 80 l/min
(21 US gpm)



3/4" L10
up to 125 l/min
(33 US gpm)



1 1/2" L10
up to 400 l/min
(105 US gpm)



2" L10
up to 800 l/min
(211 US gpm)



2 1/2" L10
up to 1400 l/min
(370 US gpm)

Characteristics

Operation	Medium	Operating pressure
solenoid,	hydraulic	1/2" - 3/4" 10–100 bar (145–1450 psi)
electrical	oil	1 1/2" - 2 1/2" 10– 59 bar (116– 856 psi)

Oil temperature range: 20°–70°C (68–158°F) - (depending on viscosity grade of oil).

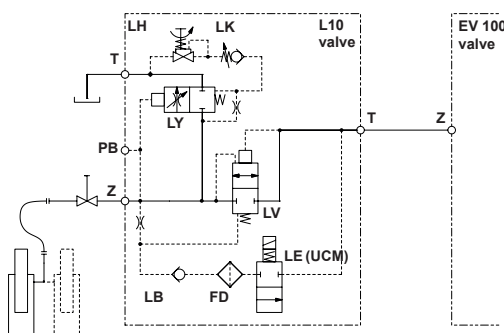
Coil insulation class ~/=: IP 68.

Description

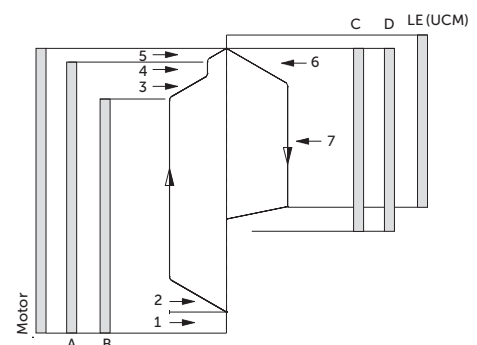
The L10 pressure lock valve is a solenoid operated check valve designed for hydraulic elevators and includes a self-closing manual lowering valve. Its purpose is to allow free flow of oil from the pump unit to the cylinder for upward travel and to prevent flow in the reverse direction from the cylinder to pump until an electrical signal is given to the solenoid.

The L10 can be mounted in any position without causing any operational problems. Installed in the main cylinder line directly adjacent to the main elevator control valve, the L10 can be employed as a safety back up valve to the down system of the main control valve to prevent unintended down movement of the elevator should an electrical or mechanical malfunction occur in the main control valve (UCM case).

Hydraulic circuit



Electrical sequence





L20

Characteristics

Operation	Medium	Operating pressure	Flow rate
solenoid, electrical	hydraulic oil	8-100 bar (116-1450 psi)	Min: 10 l/min (2.6 US gpm) Max: 125 l/min (211 US gpm)

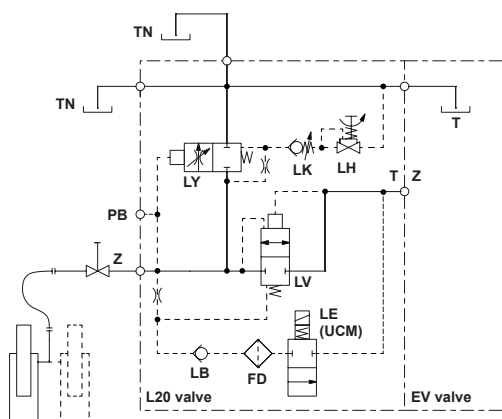
Oil temperature range: 20°-70°C (68-158°F) - (depending on viscosity grade of oil).

Coil insulation class ~/=: IP 68.

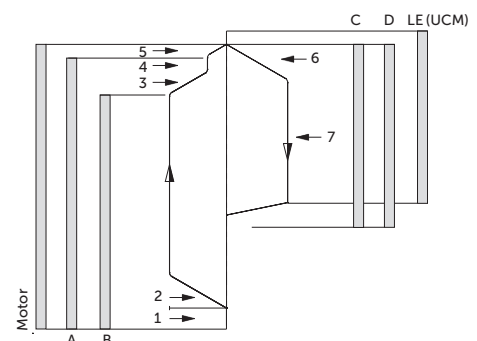
Description

The L20 has been designed to fit in all types of Blain 3/4" series of valves without the need to change any existing piping and thus is ideal for renovation projects. The L20 can be either ordered pre-assembled with a new Blain control valve or alternatively ordered as an upgrade to make an existing Blain valve compliant to european safety standards (EN 81-20/50 unintended car movement - UCM). A separate tank connection is required from L20 in case of renovation, however for a new factory assembled valve, there is no need for a separate tank connection. As no extra fittings and adapters are required, the size of the complete unit remains compact. This also results in further savings and considerable less installation time. It's an easy to implement, plug and play system.

Hydraulic circuit



Electrical sequence





3/4" iL10-S



1.5"-2.0" iL10-S

Characteristics

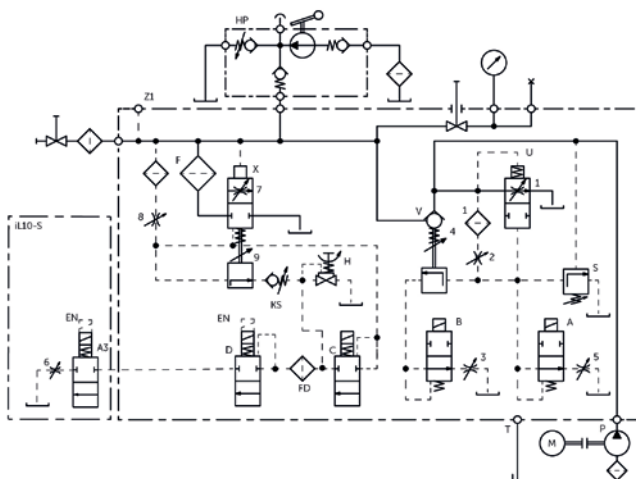
Operation	Medium	Size	Operating pressure	Flow
solenoid,	hydraulic	3/4"	10 – 95 bar	up to 125 l/min
electrical	oil	1½"-2"	8 – 75 bar	up to 800 l/min
		2½"	8 – 75bar	up to 1530 l/min

Oil temperature range: 10°C to 55°C (50°F – 131°F) – Depending on the viscosity grade of the oil. Coil insulation class ~/= : IP68

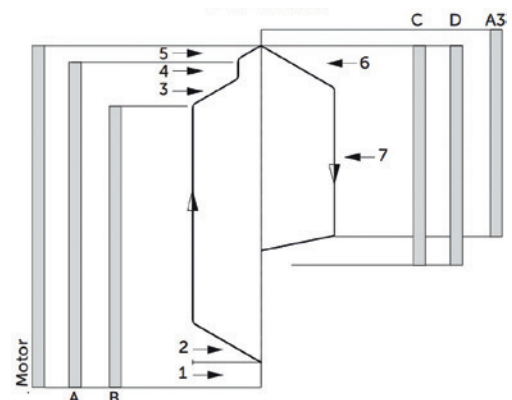
Description

The iL10-S is an advanced solenoid-operated safety lock valve engineered specifically for hydraulic elevators, offering protection against Unintended Car Movement (UCM). It is compatible with all EV-type Blain valves, including EV4, and EV40 assemblies, across a wide range of sizes from 3/4" to 2.5". The iL10-S can be quickly retrofitted to existing control valves without the need for additional adapters or connectors, resulting in significant savings in labor, materials, and weight. Its compact and simple design ensures minimal impact on system layout while enabling seamless integration with elevator controllers for real-time automatic monitoring. The iL10-S does not add extra pressure loss to the hydraulic line, thereby saving energy and preventing unnecessary heat generation. The superior features offered by the iL10-S make it not only a cost-effective and simple solution for modernizing existing power units, but also an ideal choice for enhancing elevator safety, reliability, and operational efficiency.

Hydraulic Circuit



Electrical sequence





MRL-H

Characteristics

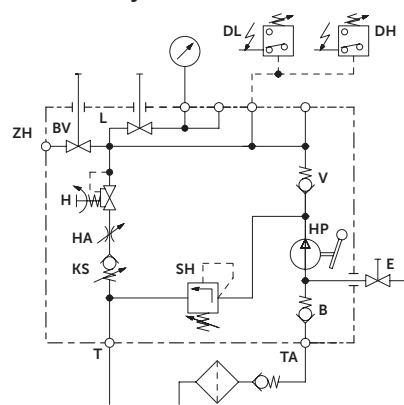
Operation	Medium	Operating pressure
manual	hydraulic oil	0–100 bar (0–1450 psi)

Oil temperature range: 20°–70°C (68–158°F) - (depending on viscosity grade of oil).

Description

The MRL-H has been designed for servicing and rescuing operations of machine room-less (MRL) hydraulic elevators remotely by having easy outside access, without needing to be in the pit. Many functional valves such as self-closing manual lowering valve, hand pump, slack rope valve, pressure relief valve, manual lowering speed adjustment, ball valve as well as a manometer have been added to a compact body. MRL-H can be located up to 6 metre (19 feet) away and 5 metre (16 feet) high from the main power unit at a convenient location for easy access. MRL-H can be optionally delivered with pipes and necessary accessories upon request.

Hydraulic circuit





A - A



E - D

Characteristics

Type AA - Female threads / Type ED - Swivel nut

Typ	size	Q max.	P max.
B3	1½"/2"	800 l/min (211 US gpm)	100 bar (1450 psi)
B5	2½"	1600 l/min (423 US gpm)	70 bar (1015 psi)

Connection possibility: 1", 1¼", 1½", 2" & 2½" - M36x2, M45x2, M52x2, M65x2 & M78x2

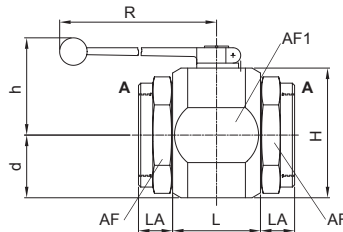
Description

The full bore ball valve provides full passage and thus causes less friction. It is universally applicable and its housing is made out of aluminium and steel.

A

Typ	A	AF*	(BSP)		(NPT)	
			Typ No.	LA	Typ No.	LA
B3	1"	70	B3G1	19	B3N1	28
	1¼"	70	B3G1.25	21	B3N1.25	28
	1½"	70	B3G1.5	24	B3N1.5	34
	2"	70	B3G2	30	B3N2	34
B5	2"	95	B5G2	31	B5N2	31
	2½"	95	B5G2.5	31	B5N2.5	35

Standard



Dimensions

Typ	DN	L	H	AF1	d	h	R
B3	38	65	90	86	43	70	240
B5	55	80	118	114	57	82	280

DN = Ø Inside

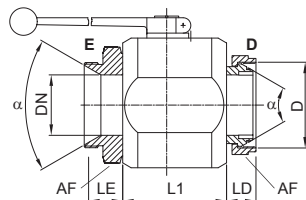
E

DIN 2353 (24°)

DIN 3863 (60°)

Typ	E	α	LE	*AF	Typ No.
B3	M36x2	24°	24.5	70	B3E36
	M45x2	24°	26.5	70	B3E45
	M52x2	24°	26.5	70	B3E52
	M65x2	60°	27	70	B3E65
B5	M78x2	60°	35	95	B5E78

Option



D

DIN 2353 (24°)

DIN 3863 (60°)

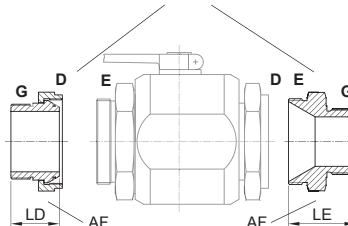
Typ	D	α	L1	LD	*AF	Typ No.
B3	M52x2	24°	66	35	60	D52
	M65x2	60°	66	25	75	D65
B5	M78x2	60°	94	24	90	D78

Adaptor GD

Typ	D	B	*AF	LD	Typ No.
B3	M65x2	G1½"	70	47	GD65.G1.5
	M65x2	Ø57 Weld	70	45	WD65.57
B5	M78x2	G2"	90	48	GD78.G2
	M78x2	Ø70 Weld	90	44	WD78.70

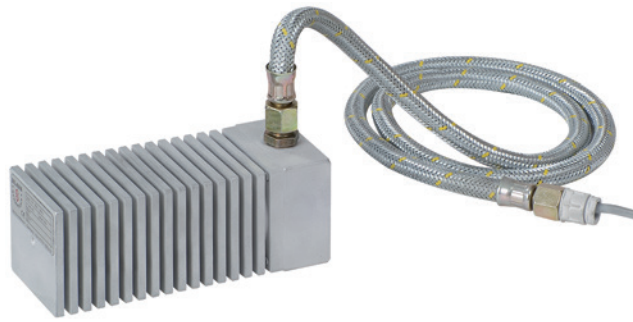
*AF-Across Flats

Adaptors



Adaptor GE

Size	E	B	*AF	LE	Typ No.
B3	M52x2	G1"	70	55	GE52.G1
	M52x2	G1 ¼"	70	54	GE52.G1.25
	M52x2	G1 ½"	70	52	GE52.G1.5
	M52x2	G2"	70	60	GE52.G2
	M65x2	G1 ½"	70	60	GE65.G1.5
B5	M65x2	G2"	70	52	GE65.G2
	M78x2	G2"	80	59	GE78.G2
	M78x2	G2 ½"	80	59	GE78.G2.5
	M78x2	NPT2 ½"	80	63	GE78.N2.5



TH

Characteristics

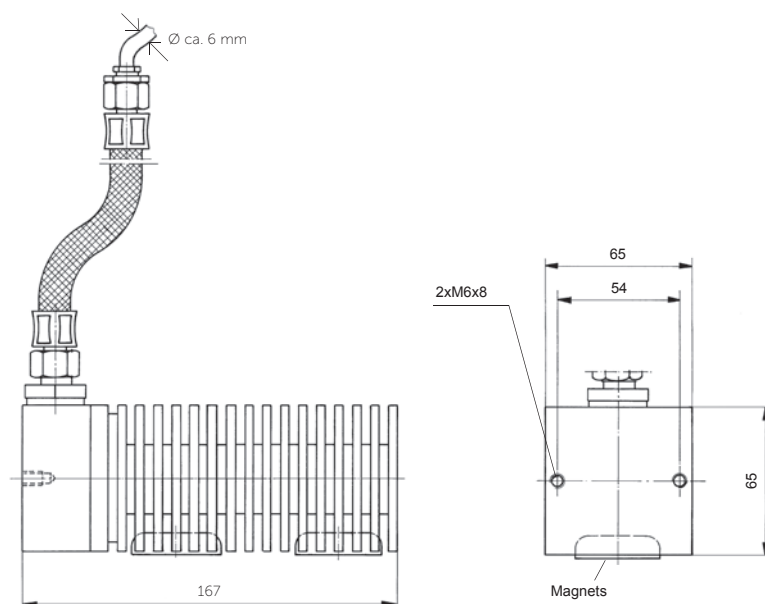
Operation	Medium	Supply	Power rating
electrical	hydraulic oil	230 VAC, 110 VAC	250 W

Description

The TH tank heaters are intended primarily for applications in hydraulic control systems for machine tools, presses, hydraulic elevators, servo systems, etc. where overnight conditions or periods of non-operation causes the temperature of the hydraulic fluid to fall below desirable levels.

The heater is designed to maintain up to approximately 500 litres (130 US gals) of oil in an unheated room at a temperature of +20 °C to +25 °C (68 °F to 77 °F). Through the large heat dissipation area of the housing, the heaters surface temperature remains under +50 °C (120 °F) and thereby avoids oxidation or premature aging of the oil. The built-in thermostat switches the heating element ON at an oil temperature of approximately +20 °C (68 °F) and OFF again when the oil temperature has risen to approximately +25 °C (77 °F).

Should the heater in an unsubmerged state be exposed to an ambient temperature of under 20 °C (68 °F), it will switch ON for a short period before switching OFF again as heat is conducted through the housing to the thermostat. Under this condition, the hottest surface temperature of the heater would not exceed 90 °C (190 °F).





H11



H12

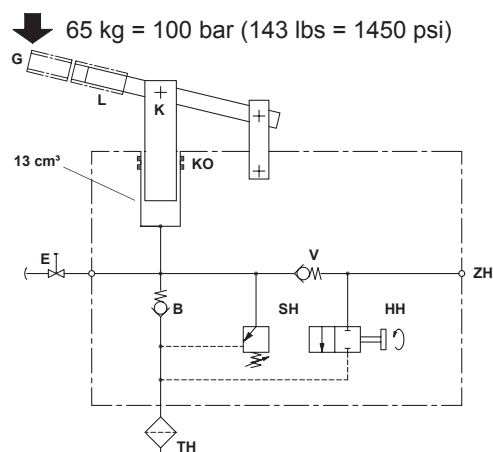
Characteristics

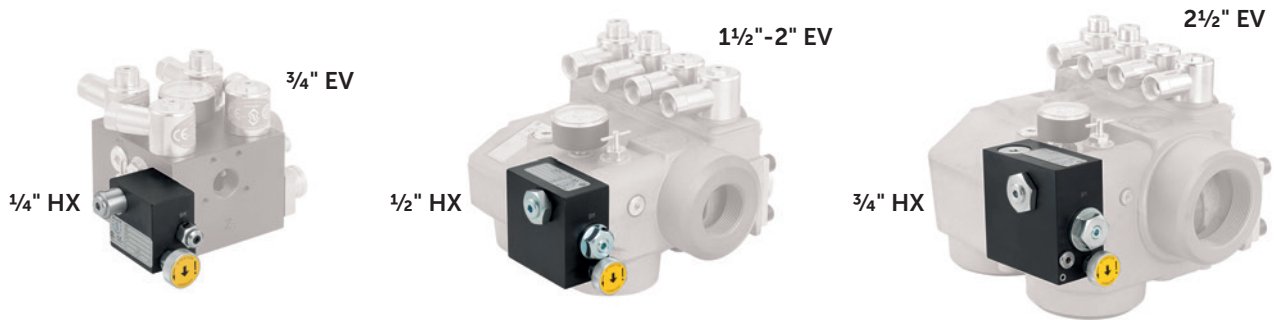
Operation	Medium	Operating pressure
manual	hydraulic oil	150 bar (2175 psi)

Description

The H11 and the H12 hand pumps are for applications with hydraulic lifting or pressing equipment, for emergency operation of hydraulic elevators and for the pressure testing of hydraulic systems in general. The H11 is constructed for side mounting. The H12 is fitted with a base plate for standalone application.

The built-in pressure relief valve should be adjusted to prevent unintentional high pressure being applied to the system. A built-in manual valve for releasing pressure from the system is available as an option.





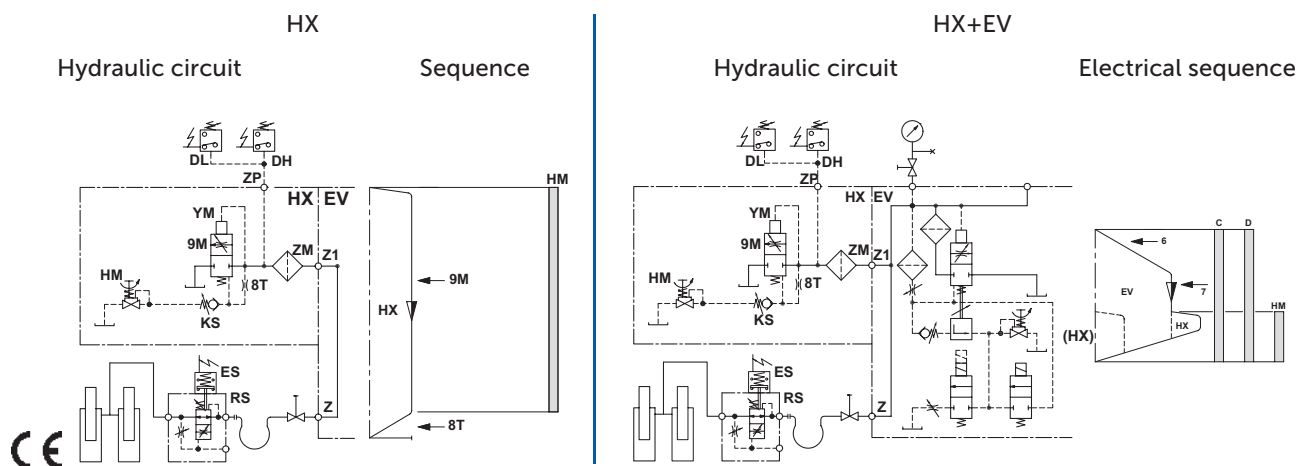
Characteristics

Operation	Medium	Operating pressure	Flow rate
manual	hydraulic oil	8–100 bar (116–1450 psi)	5–880 l/min (1.3–211 US gpm) depends on size and pressure

Oil temperature range: 20°–70°C (68–158°F) - (depending on viscosity grade of oil).

Description

The HX are manually operated down valves, adjustable in their down speed. They close automatically upon release. They can be used for emergency manual lowering or in combination with the EV down valve to achieve an overspeed of the elevator for testing the pipe rupture valve.





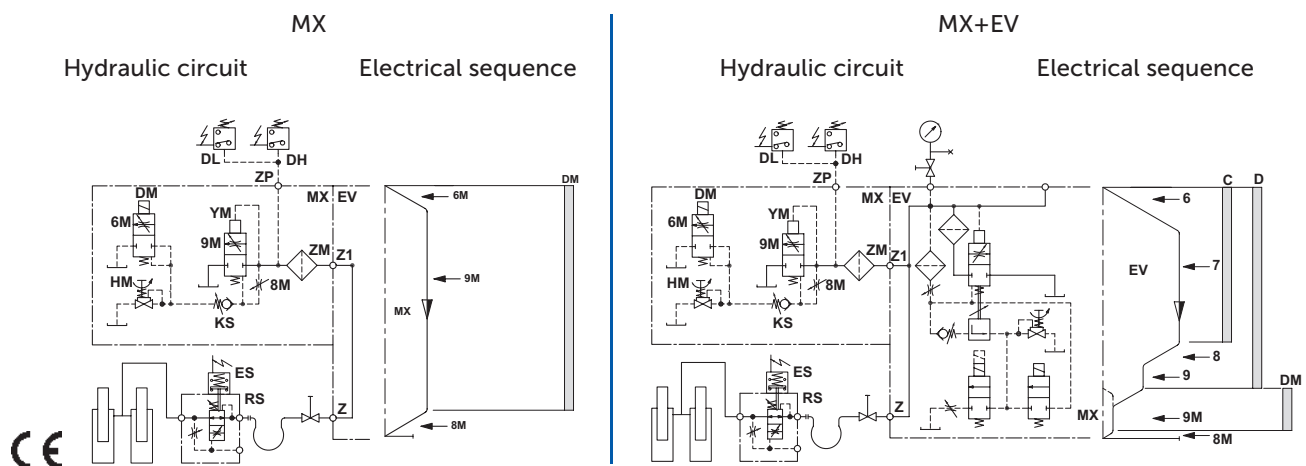
Characteristics

Operation	Medium	Operating pressure	Flow rate
solenoid, electrical	hydraulic oil	8–100 bar (116–1450 psi)	5–880 l/min (1.3–211 US gpm) depends on size and pressure

Oil temperature range: 20°–70°C (68–158°F) - (depending on viscosity grade of oil).
Coil insulation class ~/=: IP 68.

Description

The MX are solenoid operated down valves, adjustable in their acceleration, down speed and deceleration. They can be used for the revision or inspection travel of the elevator or as a particularly slow down speed valve in addition to the two down speeds of the EV valve to obtain extremely exact floor stops.





EN

Characteristics

Emergency supply	Main supply
12 VDC (2 A)	24 VDC, 48 VDC, 110 VDC, 180 VDC, 110 VAC, 230 VAC
24 VDC (1.1 A)	24 VDC, 48 VDC, 110 VDC, 180 VDC, 110 VAC, 230 VAC

Description

Should there be an interruption of the main power to the elevator, the emergency lowering coil EN, fed by an emergency 12 VDC or 24 VDC supply, enables a command to be given from the car or elsewhere to lower the car to the floor below. When ordering please state main and emergency voltages.


1/2" KSB

up to 80 l/min (21 US gpm)

3/4" KSB

up to 125 l/min (33 US gpm)

1 1/2" KSB

up to 400 l/min (105 US gpm)


2" KSB

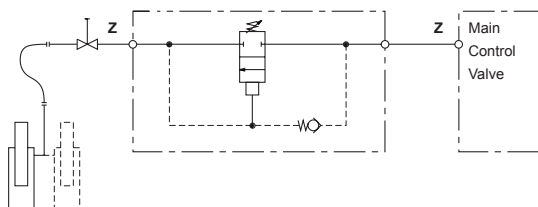
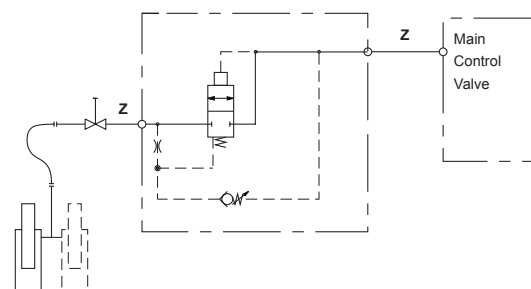
up to 800 l/min (211 US gpm)

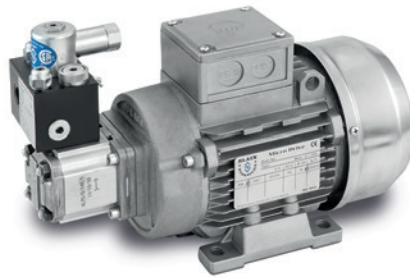
Characteristics

Operation	Medium	Operating pressure
—	hydraulic oil	10-100 bar (145-1450 psi)

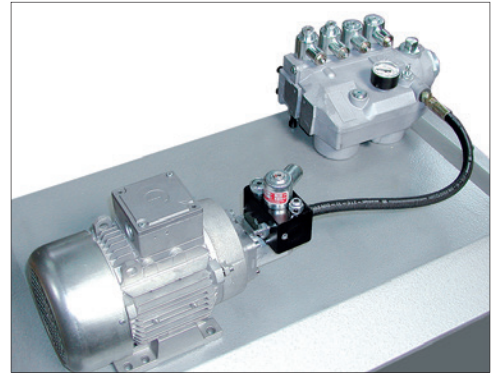
Description

Slack rope valve for separate installation. It prevents the slack rope condition caused by the lowering of the ram when the car is suspended in the safeties or resting on the buffers.

1/2" KSB

3/4" & 1 1/2" & 2" KSB




MD



Characteristics

Operation	Medium	Operating pressure	Flow rate
solenoid, electrical	hydraulic oil	Max: 130 bar (1885 psi)	Min: 1 l/min (0.3 US gpm) Max: 24 l/min (6.3 US gpm)

Oil temperature range: 20°-70°C (68-158°F) - (depending on viscosity grade of oil).

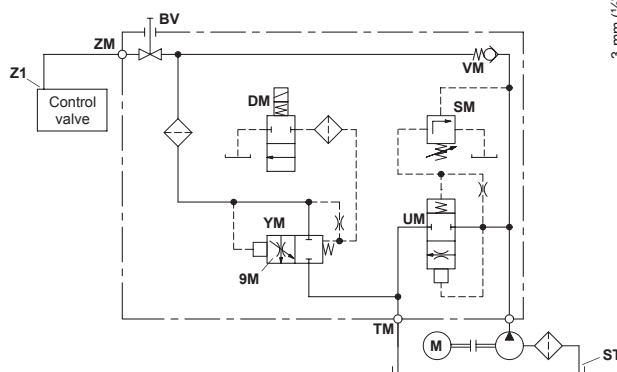
Coil insulation class ~/= IP 68.

Description

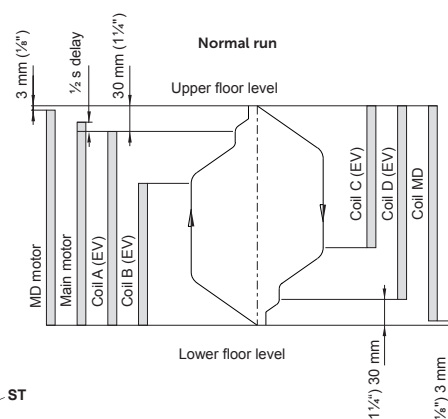
The Blain Micro Drive for hydraulic elevators consists of a small motor, pump and valve unit in one assembly. Exact floor stops and re-levelling operations are achieved with low electrical power requirement, low noise levels and no unnecessary heating of the oil.

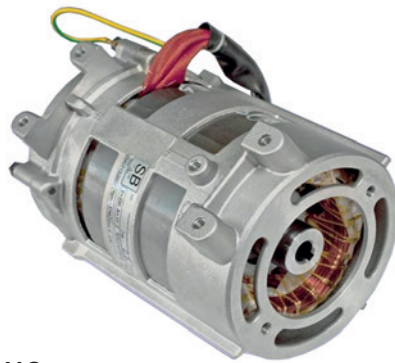
The MD unit is mounted on or under the cover of the main hydraulic power unit, using the same oil source. It can also be used to slowly move the car independently of the main drive during installation or in an emergency.

Hydraulic circuit



Electrical sequence





MO

Characteristics

Type SB mini lift single phase (50 or 60 Hz) or three phase (50 Hz):

- Type SB Mini oil	(50 or 60 Hz):	1.5-3.3 kW	(2 - 4.5 Hp)
- Type SB 150-A	(50 or 60 Hz):	4.7-22 kW	(6.5 - 30 Hp)
- Type SB 150-B	(50 or 60 Hz):	12.5-22 kW	(17 - 30 Hp)
- Type SB 200	(50 or 60 Hz):	29.4-44.1 kW	(40 - 60 Hp)
- Type SB 250	(50 or 60 Hz):	51.5-73.5 kW	(70 - 100 Hp)

Description

wSB Motori submersible single and 3 phase motors are specifically designed for immersion in oil to work with submersible screw pumps and meet the requirements of low noise level and high efficiency in hydraulic lifts.

Advantages

Submersible motors offer unique advantages like:

1. Silent operation (by virtue of being submerged in oil inside the tank)
2. Direct coupling with submersible pump (no need of bell housing and coupling)
3. Very compact size and light weight (compared to big and heavy external motors)
4. Aesthetic and compact power unit design



UOP

Characteristics

Data at 50 cSt, 2750 rpm and 40 bar: PA european standard

Temperature

0 to 100 °C (32-212 °F)

Pressure max.

60 bar (870 psi) continuous

Flow rate

22.6-873 l/min (6-230 US gpm)

Data at 50 cSt, 2750 rpm and 40 bar: without bell housing

Temperature

0 to 120 °C (32-248 °F)

Pressure max.

75 bar (1087 psi) continuous

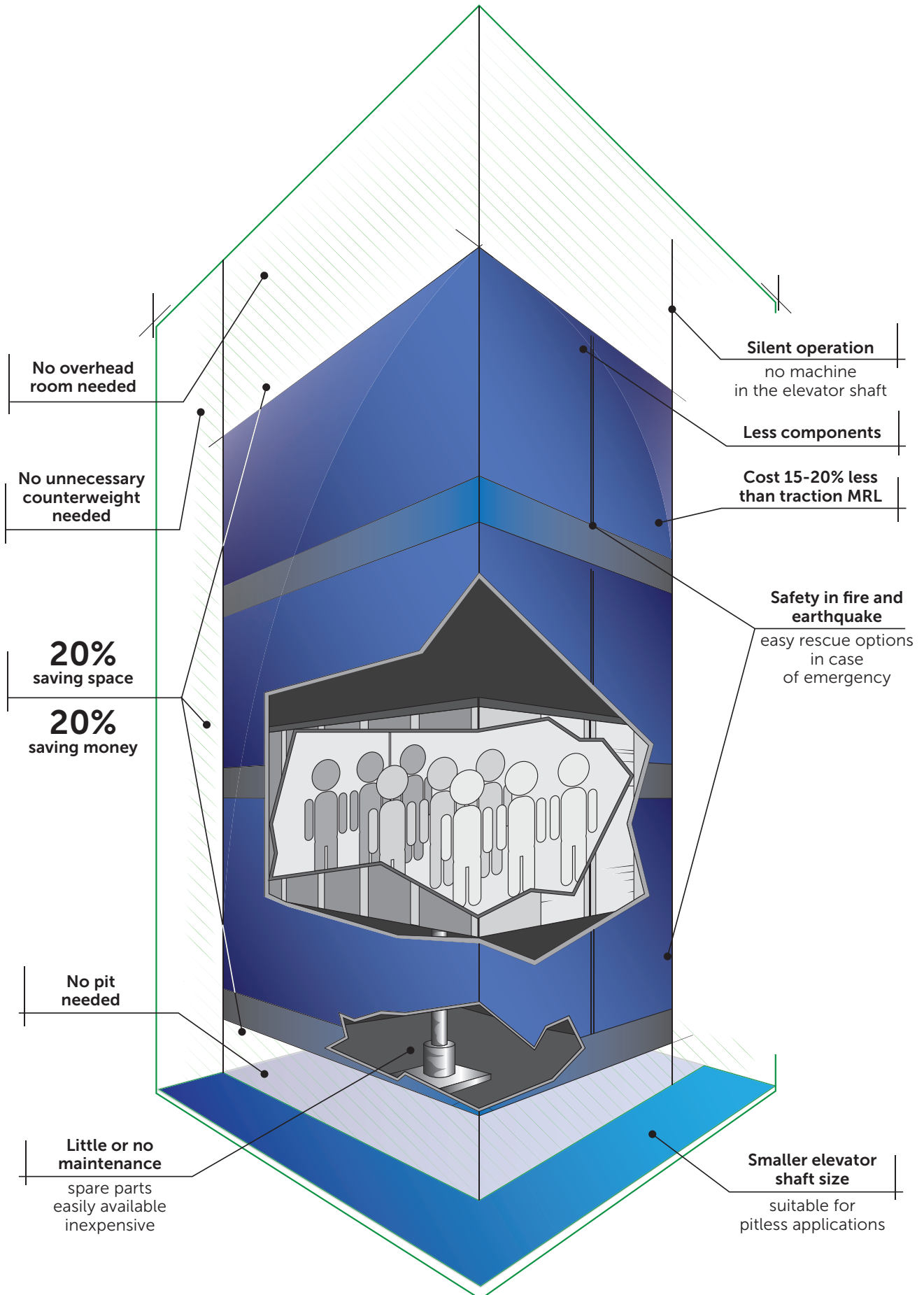
Flow rate

8-26 l/min (2-6.9 US gpm)

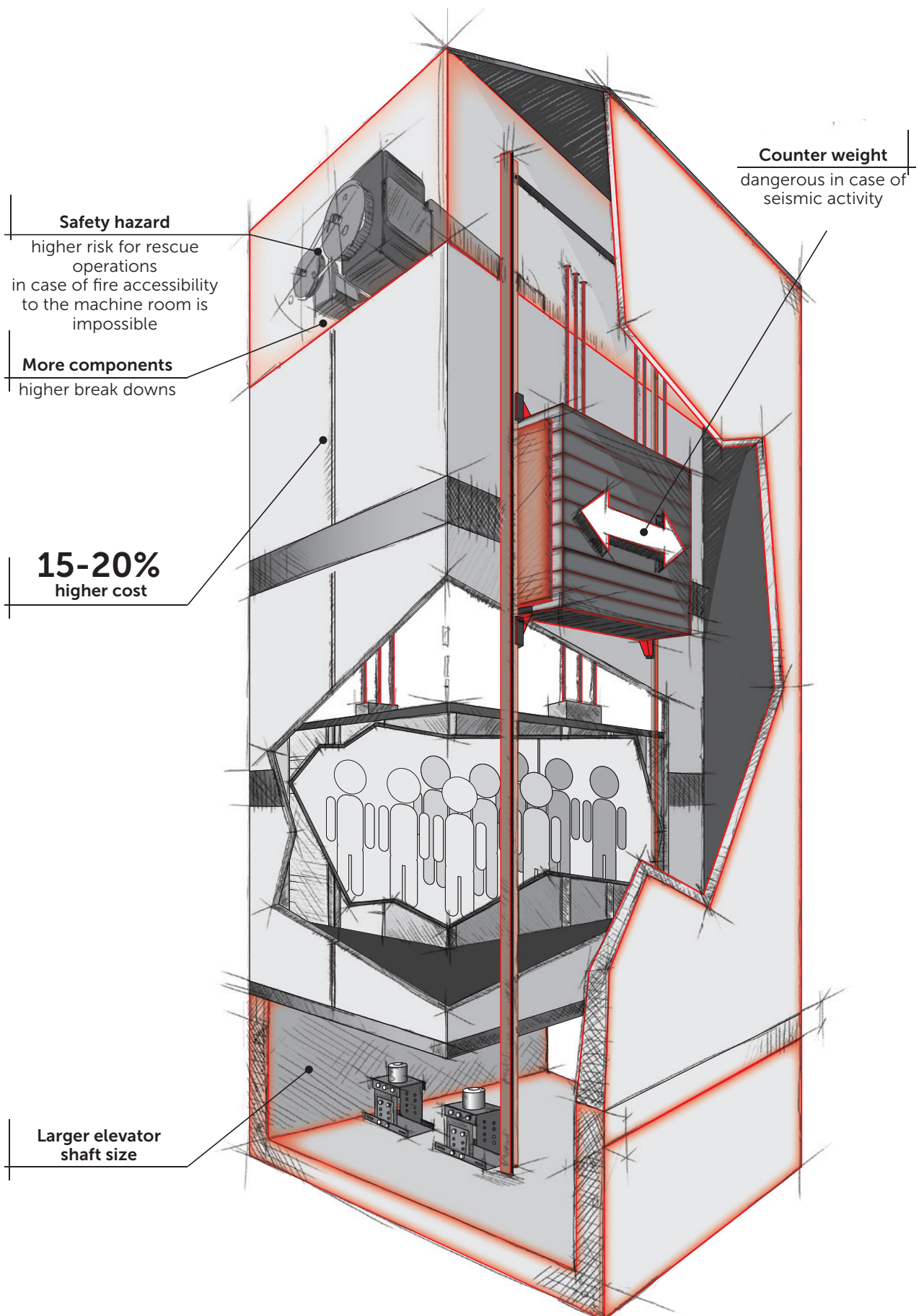
Description

Submersible screw pumps are ideal for use in hydraulic elevators due to the fact that they are silent in operation, offer good efficiency and low pulsation.

HYDRAULIC LIFTS



TRACTION/MRL LIFTS





Anja Blain CEO

Since half a century Blain Hydraulics has focused and specialized in flow control valves for hydraulic elevators. Blain is the largest supplier of elevator valves in the world with a large global footprint. At Blain safety, reliability and quality of our products are of utmost importance. As a pioneer, Blain has been building products which are above and beyond the standards. With product support in multiple languages and across different time zones, more than a million valves in operation worldwide endorse us a leading supplier of key elevator components.

At Blain, flow control is in our DNA, we don't just manufacture a valve, we engineer it.

Anja Blain (Managing Director/CEO)

BLAIN HYDRAULICS is the leading manufacturer of high quality hydraulic elevator products for five decades. Blain products have proven their safety and quality by possessing more than one third of the global market share and one million valves in operation in more than 75 countries worldwide.



A brief history of Blain Hydraulics – incorporated in 1971 by Roy W. Blain



Roy W. Blain 1932-2014

Roy W. Blain was born in Salford, Manchester, in May 1932 and studied mechanical engineering at Salford Royal Technical College in northern England.

Following two years as a sailor in the Merchant Navy, he served in the Army for a further two years. He then pursued a career in industrial hydraulics, working first in England and then in Switzerland, Spain, and the USA. Upon arriving in Germany, he founded Blain Hydraulics. Roy W. Blain was a true pioneer in the field of hydraulic lifts. A genuine gentleman and kind-hearted visionary, he worked in the field of hydraulic lift technology for more than five decades.

1971-1980 During this decade Blain Hydraulics GmbH was founded in Heilbronn. With a modest infrastructure as well as man power, elevator control valves like EV & KV started rolling out initially with 1 person and eventually with 5 people on the outskirts of Heilbronn. To cater for growing demand, the factory was relocated within Heilbronn and steadily expanded.

1981-1990 During this period, Blain adds new KV models (small lift valves) to its range, to meet demand for small lifts in commercial and private settings. In addition, the L10 pressure relief valve is introduced as an additional safety valve. This product is now known as the UCM-A3 valve. Blain obtains CSA certification for the export of goods to North America. The company expands its infrastructure to meet growing customer demand.

1991-2000 Over the course of this decade, the company has modernised its machinery to make production more cost-effective and profitable. Blain obtained ISO 9001 certification and successfully obtained type approval for its pipe rupture valves in accordance with EU directives. The company's product range was expanded. Blain introduces the SEV electronic servo valve and other products, such as the MD fine levelling drive. Additional equipment, such as ball valves, is also added to the product range.

2001-2010 Blain is the first company worldwide to bring

explosion-proof MEX solenoid valves for lifts to market. The company becomes the world's largest manufacturer of lift control valves in terms of production capacity and installations. Additionally, new pipe rupture valves were introduced during this decade.

2011-2020 Blain launches the EV4 (VVVF control valve) in collaboration with YASKAWA as part of a joint venture. Exports of Blain products reach a new record high. In 2015, Blain expanded its presence in India by setting up Blain India. A partnership with Brazilian company DAIKEN ELEVADORES is formed, enabling Blain to increase its presence in the hydraulic lift market in South America. In summer 2017, Blain Hydraulics introduces the L20 as a new UCM-A3 valve. This enables existing installations to be modernised with less effort and at lower cost.

In 2018, Blain's presence is expanded to the Middle East and Africa with the establishment of a location in Turkey. With the founding of Blain Hydraulics USA in 2018, the company sends a strong signal of customer proximity in North America. The direct presence in the North American market enables faster, more personal and more efficient customer service. In 2019, Blain presents the next generation of smart valves, which offer users easy installation and monitoring via smartphone or tablet. This new product line consists of the SEV servo electronic valve and the EV40 VVVF valve. With these products, Blain Hydraulics is the first company to introduce smart technology to the hydraulic lift industry, once again demonstrating Blain Hydraulics' innovative strength.

2021-today Blain Hydraulics is investing in the sustainable design of its production site. The company's roofs are being renovated and fitted with photovoltaic systems. This enables the company to generate a large proportion of its own electricity. The workstations in the assembly department are being designed ergonomically. With the purchase of automatic storage systems more space at the site is created. In 2024, the product family will be expanded to include another safety component. The new iL 10-S is also TÜV tested and approved. While it differs visually from its predecessor, the iL 10, it has the same function. The company now has an international customer base spanning more than 75 countries. Blain currently employs around 120 people from over 14 countries.





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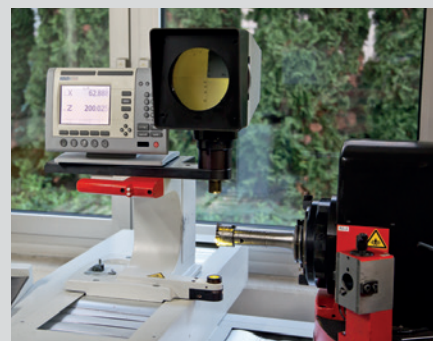
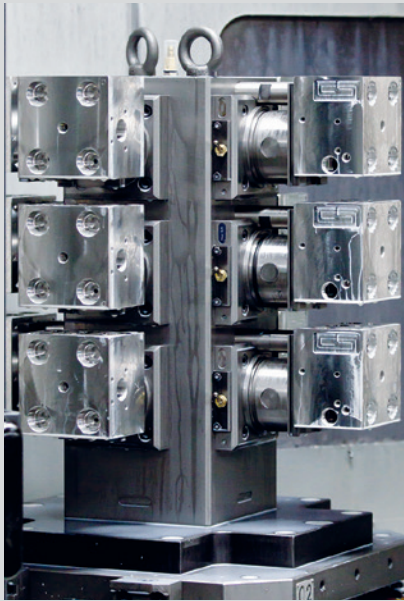


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blain.de

BLAIN HYDRAULICS

Designer and Manufacturer of the highest quality control valves & safety components for hydraulic elevators